Louis Wellmeyer

Data Science Graduate Student, KAIST | B.S. Computer Science | Ex-GCF

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Summary

Software and data scientist focusing on reinforcement learning in simulated environments. Passionate about data-heavy, large-scale system design, with a total of **4 years of experience** across AI engineering and agent orchestration, data engineering, automation, fullstack development, teaching, and technical project management, spanning research, industry, and international organizations, among which the Green Climate Fund under UNFCCC.

Education

Korea Advanced Institute of Science and Technology (KAIST) *2026 – 2028 (expected)*

M.S. in Data Science, Graduate School of Data Science

- Research focus: reinforcement learning for manufacturing scheduling, System Analytics Lab (advisor: Prof. Hayong Shin)
- Member of Microrobot Research Club, focusing on Sim2Real (MuJoCo, Nvidia Omniverse IsaacSim)

RWTH Aachen University *2021 – 2025*

B.S. in Computer Science

Goethe-Gymnasium Ibbenbüren *2013 – 2021*

High-School Diploma (Abitur), Major: Mathematics & Computer Science

Experience

Green Climate Fund (UNFCCC) - Technical Business Partner *Jul 2025 – Jul 2026*

Songdo International Business District, South Korea

- Led **20+** cross-departmental IT initiatives in parallel, translating organizational needs into technical requirements and aligning solutions with business objectives across stakeholders and global technology vendors
- Planned and executed GCF's first-ever Tech & AI Exhibition (**4+ months**, **8** departments, **4** booths, **16+** initiatives) engaging Board members, accredited entities, and country representatives on AI governance and digital equity, with **3+** external vendor coordination including **Microsoft**
- Designed data pipelines and analytical models for climate-related forecasting and country vulnerability assessments (**Microsoft Fabric**, **Spark**, **Kerchunk/NetCDF** indexing, **NLP**)
- Built an AI-assisted longlisting solution, improving process efficiency by over **3**× while maintaining evaluation quality
- Developed enterprise Finance and HR platforms (**Next.js**, **TypeScript**, **PostgreSQL**, **Azure**, **SAML/SCIM**), including a multi-year budgeting platform with approval workflows and **SSO**/DocuSign integration

Teutoburger Ölmühle GmbH - Fullstack Developer *Apr 2025 – May 2025*

Ibbenbüren, Germany

- Redesigned and rebuilt the company's internal intranet from scratch (**Angular**, **FastAPI**) after reverse-engineering a legacy **PHP** multi-page-application codebase
- Refactored the data pipeline to integrate directly with **Microsoft NAV** databases, enabling real-time data feedback
- Introduced **GitHub** for version control and set up a CI/CD pipeline via **GitHub Actions**

RWTH Aachen University - Learning Technologies Research Group *Oct 2024 – May 2025*

Aachen, Germany

- *Data Engineering Research Assistant* (Feb – May 2025): designed and deployed containerized analytic engines and RESTful APIs (**Python/Flask**, **Docker**) for real-time analysis of student data across multiple German universities; sped up the pipeline up to **5**× via **multiprocessing**-based load balancing
- *Data Science Teaching Assistant* (Oct 2024 – Apr 2025): instructed and mentored international cohorts of up to **40** students on **Python** and **Pandas**-based data analysis, with individualized code review and feedback

Goethe-Gymnasium Ibbenbüren - Robotics Club President *Jul 2019 – Jul 2021*

- Led weekly robotics sessions for up to **20** students, teaching programming and engineering fundamentals through to autonomous robotic systems
- Coordinated national-scale robotics competitions involving schools across Germany

Selected Projects

ThinkSlides - Multi-Agent Presentation Generation *Next.js, Go, Python*
ReAct agent system generating branded PowerPoint presentations via a streaming 5-service architecture

GreenLense - Agentic Product-Insights API *Python, OpenAI* | Code
Hackathon finalist, Hack Seoul 2025 (AngelHack × coupang Coupang)

NASA Turbofan Remaining-Useful-Life Prediction *Python* | Code
End-to-end data science pipeline predicting engine maintenance needs and remaining useful life

wellm.nvim - Neovim LLM Integration Plugin *Lua* | Code
Async streaming LLM integration with native function-calling; reduces context size by ∼**90%** via symbol-outline injection

SpeechFunctionCaller - Speech-to-Browser-Action Framework *TypeScript, Java, Azure OpenAI* | Code
Bachelor's thesis project enabling hands-free voice control of web applications via LLM-driven function calling

Technical Skills

Languages	Python, Go, TypeScript, SQL, C++
Data / ML	Pandas, Scikit-learn, Apache Spark, Databricks, PyTorch, RAG, MCP, MLOps
Backend	FastAPI, REST APIs, PostgreSQL, Redis, MongoDB, pgvector
Infrastructure	Docker, Kubernetes, Git, Azure, GitHub Actions
Environment	Linux, Neovim, Claude Code

Academic Writing

- **Enabling Voice Input for Data Entry through LLMs in MontiGem**, Bachelor's Thesis, RWTH Aachen, 2024/25 | PDF
- **Pursuing Ideal Regression Testing for HPC: A Comparative Framework Analysis**, Seminar, RWTH Aachen, 2023/24 | PDF
- **Simulating Stiffness of Virtual Objects with Haptic Jamming Devices: An Overview**, Seminar, RWTH Aachen, 2022 | PDF

Languages & Certifications

Languages: German (native), English (fluent), Korean (intermediate)

Certifications: Introduction to Statistics (Stanford) | Machine Learning Operations Engineer Associate (Microsoft) | Google Project Management (Google) | IBM Data Science (IBM)